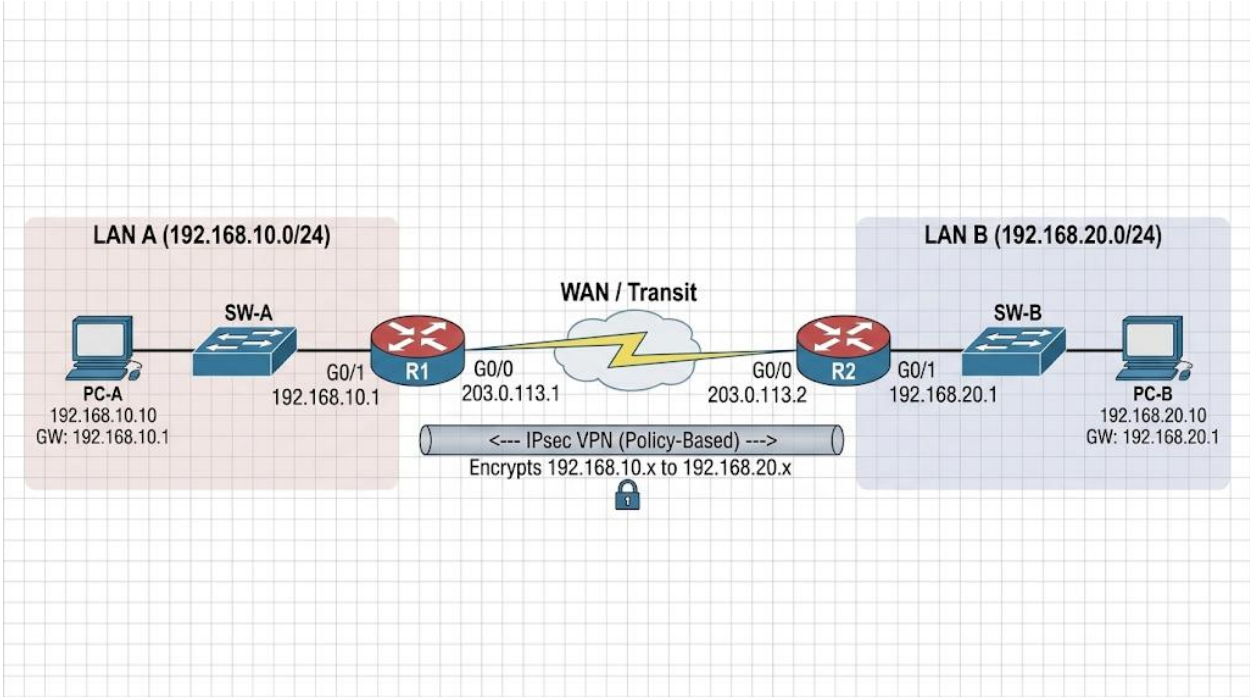
IPsec VPN



Interface and Addressing Table

Node / Device	Interface	IP Address	Subnet Mask	Purpose
R1	GigabitEthernet0/0	203.0.113.1	255.255.255.252	WAN / VPN peer
R1	GigabitEthernet0/1	192.168.10.1	255.255.255.0	LAN A default gateway
R2	GigabitEthernet0/0	203.0.113.2	255.255.255.252	WAN / VPN peer
R2	GigabitEthernet0/1	192.168.20.1	255.255.255.0	LAN B default gateway

Node / Device	Interface	IP Address	Subnet Mask	Purpose
PC-A	NIC	192.168.10.10	255.255.255.0	Host behind R1 (GW: 192.168.10.1)
PC-B	NIC	192.168.20.10	255.255.255.0	Host behind R2 (GW: 192.168.20.1)

Router 1 (R1) Complete Configuration

```
configure terminal

 1. Physical Interfaces
interface GigabitEthernet0/1
 description LAN-A
 ip address 192.168.10.1 255.255.255.0
 no shutdown
 exit
```

```
interface GigabitEthernet0/0
 description WAN-Link
 ip address 203.0.113.1 255.255.255.252
 no shutdown
 exit

 2. Phase 1: ISAKMP (IKEv1) Policy & Key
crypto isakmp policy 10
 encr aes 256
 hash sha
 authentication pre-share
 group 14
 lifetime 86400
 exit
crypto isakmp key LabOnly-ChangeMe address 203.0.113.2

 3. Phase 2: IPsec Transform Set
crypto ipsec transform-set TS-AES-SHA esp-aes 256 esp-sha-hmac
 mode tunnel
 exit

 4. Traffic Selector (Interesting Traffic ACL)
ip access-list extended VPN-TRAFFIC
 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
 exit
```

5. Crypto Map Configuration

```
crypto map S2S-MAP 10 ipsec-isakmp
 set peer 203.0.113.2
 set transform-set TS-AES-SHA
 match address VPN-TRAFFIC
 exit
```

6. Apply Crypto Map to the WAN Interface

```
interface GigabitEthernet0/0
 crypto map S2S-MAP
 exit
```

7. Routing

Note: Policy-based IPsec intercepts traffic based on the ACL.

You still need a route to send the traffic out the WAN interface.

```
ip route 192.168.20.0 255.255.255.0 203.0.113.2
end
write memory
```

Router 2 (R2) Complete Configuration

```
configure terminal
```

1. Physical Interfaces

```
interface GigabitEthernet0/1
 description LAN-B
```

```
 ip address 192.168.20.1 255.255.255.0
 no shutdown
 exit

interface GigabitEthernet0/0
 description WAN-Link
 ip address 203.0.113.2 255.255.255.252
 no shutdown
 exit

 2. Phase 1: ISAKMP (IKEv1) Policy & Key
crypto isakmp policy 10
 encr aes 256
 hash sha
 authentication pre-share
 group 14
 lifetime 86400
 exit
crypto isakmp key LabOnly-ChangeMe address 203.0.113.1

 3. Phase 2: IPsec Transform Set
crypto ipsec transform-set TS-AES-SHA esp-aes 256 esp-sha-hmac
 mode tunnel
 exit

 4. Traffic Selector (Interesting Traffic ACL)
```

```
ip access-list extended VPN-TRAFFIC
 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
 exit

 5. Crypto Map Configuration
crypto map S2S-MAP 10 ipsec-isakmp
 set peer 203.0.113.1
 set transform-set TS-AES-SHA
 match address VPN-TRAFFIC
 exit

 6. Apply Crypto Map to the WAN Interface
interface GigabitEthernet0/0
 crypto map S2S-MAP
 exit

 7. Routing
ip route 192.168.10.0 255.255.255.0 203.0.113.1
end
write memory


show crypto ipsec sa
```